PROBLEM STATEMENT BY INKLOOM AI • UPDATED SOLUTION PROPOSAL & ROUND 2 BUILD REPORT

BrandGuard

An intelligent brand-governance platform that reviews content before it is published

Project Attribute	Submission Details
Team name	Nexus
Team members	Logeshwaran V, Muthukumar A, Mohammed Arshath, Poojitha V S
Institution / organisation	SRM Easwari Engineering College
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Public GitHub repository	https://github.com/Logesh6767/BrandGaurd
Submission round & date	Round 2 — The Build (7th Nov 2026) | Original Proposal: 3rd Nov 2026

1. Understanding of the problem statement & Round 2 Executive Summary

Organisations now publish through many hands at once: employees, agencies, freelancers and AI writing tools. Each contributor interprets the brand slightly differently, and nobody sees the whole picture before content goes live. The result is content with wrong or outdated facts, unsupported promotional claims, off-tone language, misused terminology, off-palette visuals and inaccessible design. These errors usually become visible only after publication, when the damage to credibility is already done.

> **Round 2 Build Status — Central Workflow Running End-to-End:** In Round 2, Team Nexus turned the Round 1 proposal into a complete, production-ready web application built with Python 3.12, FastAPI, SQLite3, and an Apple-HIG Single-Page Application (SPA). Opening the deployed link in an incognito window loads straight into an active workspace pre-seeded with the ABC SmartFlow Global Brand Kit v3.0 (14 verified rules), 3 live submissions (including the exact BG-001 finding from Round 1), a 1-click Role Switcher (Creator, Reviewer, Manager, Admin), and a 27-step interactive guided tour.

Root causes addressed by BrandGuard

- **No shared source of truth:** Guidelines sit in scattered PDFs, emails and people's heads instead of an owned, versioned rulebook.
- **Outdated information:** Expired specifications, prices ($19/month legacy tier) and old product names keep circulating with external agencies.
- **Ungrounded AI writing tools:** Generic LLMs write fluent copy with invented claims because they cannot verify approved company facts.
- **Manual review bottleneck:** A handful of brand and legal reviewers cannot inspect every post manually without delaying launches.
- **No feedback loop:** Mistakes are fixed one by one without tracking which rules are repeatedly dismissed because the guideline itself is unclear.

> **Our reading of the core challenge is trust.** Reviewers will rely on an automated system only if each finding is grounded in an approved source, can be explained in plain language, and can always be overruled by a person.

2. Users affected by the problem & how the live build serves them

User Persona	How the problem affects them	What BrandGuard delivers in the live Round 2 build
Content creators (marketers, social teams, writers)	Unsure of the latest rules; late rework; fear of publishing a mistake.	Instant inline highlights, cited rules, and 1-click intent-preserving rewrites that keep their message.
External agencies and freelancers	Little access to internal guidance; the most revision rounds.	Pre-screened self-check against the active Brand Kit v3.0 before submitting to internal reviewers.
Managers and team leads	Spend time chasing approvals and fixing repeat errors.	Live 6-stage workflow stepper, real-time Brand Score (0-100%), and cross-team compliance insights.
Brand, legal and compliance reviewers	Review everything manually; bottleneck; liability for wrong claims.	Pre-screened content, verified citations, and mandatory human escalation gate on high-risk dismissals.
Brand owners and administrators	Guidelines are scattered, outdated or ambiguous; no feedback on which rules fail.	Versioned Brand Knowledge Base (v1-v3.0) with live rule auto-fill editor and Ambiguous-Guideline Detector.

3. Proposed solution & Six Core Design Principles (All Active in Round 2)

- **1. Evidence-grounded:** No finding is produced without citing a rule ID, source document, owner and version, plus quoting the exact source text verified by the Evidence Validator (backend/engine.py).
- **2. Hybrid engine:** Deterministic code checks exact rules (banned terms, expired facts, WCAG 2.1 contrast, alt text, CTA links, palette distance). The AI layer (Claude API with offline heuristic fallback) handles tone and claim support.
- **3. Intent-preserving suggestions:** Each suggested rewrite makes the smallest change needed and states explicitly what creator purpose was preserved.
- **4. Human control & Step 5 Escalation:** Creators accept, edit or dismiss with a written reason. Dismissing a HIGH-severity finding automatically escalates it to a reviewer instead of clearing it.
- **5. Confidentiality and transparency:** Text inside [[double brackets]] is stripped locally before any external AI call, and every run is logged with a SHA-256 payload hash in the Audit & AI Trust Log.
- **6. Learning loop:** Rules with high dismissal rates (>= 40%) are flagged in the Governance Insights dashboard with a 1-click rule clarification flow.

Flagship Round 1 Example Finding — Live in Submission #1 (BG-001)

Field	Live Verified Output in BrandGuard Review Studio
Issue / severity	BG-001 | HIGH (Exact Fact Contradiction & Unsupported Guarantee)
Detected text	"ABC SmartFlow supports 20 languages and guarantees 100% growth."
Rule and source	Fact F-12 "Supports 5 languages" (Product Specification v3, owner: Product Team). Claims Policy rule C-4: no unsupported guarantees.
Why flagged	The language count (20) contradicts the approved fact (5 languages), and the 100% growth guarantee is an unsupported promotional promise.
Suggested fix	"ABC SmartFlow supports 5 languages and helps businesses improve operational efficiency." Preserves: product benefit and positive launch tone.
Confidence / status	High (98% — exact fact mismatch) | Verified Grounding | Needs human verification

4. Complete 9-Step Product Workflow (Implemented & Running End to End)

Step	What happens in the live Round 2 product	Role	Status
1. Set up brand kit	Manage guidelines, terminology, approved facts (with owner and expiry), claims, palette and accessibility rules in Brand Kit v3.0. Typing or clicking any existing Rule ID auto-fills all fields.	Admin / Owner	COMPLETED
2. Create content	Submit text, visual HEX palette, or an image with alt text and CTA link text. Text inside [[double brackets]] is marked confidential.	Creator	COMPLETED
3. Automated review	Redacts [[confidential]] spans first, runs deterministic + AI/heuristic checks, and verifies every finding through the Evidence Validator.	System	COMPLETED
4. Creator revision	1-click Accept Fix (saves a new numbered version v2, v3... with a word-level diff), custom edit, comment, or dismiss with a written reason.	Creator	COMPLETED
5. Escalation safety net	A HIGH-severity (factual or legal) finding dismissed by a creator automatically enters ESCALATED (Amber State) instead of being cleared.	System	COMPLETED
6. Submit for approval	Gate unlocks only when 0 OPEN findings remain and the text has been checked since its last change.	Creator	COMPLETED
7. Human approval	Reviewer/Manager must resolve any ESCALATED findings (Override & Clear vs. Uphold Rule & Require Fix) before approving, requesting changes, or rejecting.	Reviewer / Manager	COMPLETED
8. Publish-ready record	Approved content generates a tamper-evident Compliance Certificate with SHA-256 content hash, version history, word-level diffs, comments, and AI log.	System	COMPLETED
9. Governance insights	Tracks recurring violations by rule and team, average revisions to approval, and flags high-dismissal rules (>= 40%) for 1-click clarification.	Manager / Admin	COMPLETED

5. Technology Stack & Implementation Details

Layer	Implemented Choice in Round 2	Why & How It Fits Together
Frontend	HTML5, CSS3 (Apple HIG / macOS Sequoia design theme), Vanilla ES6 JavaScript SPA (app.js, tour.js)	Zero build-step fragility; split-pane Review Studio, live WCAG contrast preview, word-level diff viewer, and 27-step interactive guided tour.
Backend	Python 3.12 with FastAPI, Pydantic v2, Uvicorn	Serves both the REST governance API (/api/*) and the SPA frontend from a single unified deployment.
Database	SQLite3 with 9 normalized relational tables (backend/database.py)	Stores users, versioned brand kits, 14 canonical rules, submissions, revisions, findings, comments, AI disclosure logs, and audit events.
AI & Fallback	Anthropic Claude API (claude-3-5-haiku-latest) + deterministic offline heuristic judge	Evaluates nuanced tone (G-01) and claim support (C-02) on redacted text; works 100% reliably without external API keys via built-in fallback.

Layer	Implemented Choice in Round 2	Why & How It Fits Together
Deterministic Checks	Regex boundary matchers, ISO expiry checks, WCAG 2.1 luminance/contrast math, RGB Delta-E distance, Flesch-Kincaid grade	Executes in <15ms with 100% precision and zero hallucination risk for factual, visual, and accessibility rules.
Security & Privacy	PBKDF2-HMAC-SHA256 (100k iterations), HMAC-SHA256 bearer tokens, server-side RBAC, [[confidential]] redactor	Protects pre-release metrics from external AI exposure and enforces role permissions on every API action.

6. What Changed from the Original Round 1 Proposal, and Why

- **1. Added 1-Click Judge Persona Switcher & Pre-Seeded Incognito Workspace:** Alongside formal PBKDF2/HMAC token login, we added a 1-click Role Switcher in the sidebar (Creator, Reviewer, Manager, Admin) that issues real signed tokens on the fly so judges testing in an incognito window can experience every role gate immediately without login friction.
- **2. Added 27-Step Interactive Spotlight Tour & Auto-Dismissing Helper Callout:** To ensure anyone who did not build the product can immediately understand all 9 workflow steps, we built a custom guided tour engine (tour.js) with choreographed workspace transitions and role-guide drawers.
- **3. Client-Side HTML5 Canvas Dominant Color Extraction:** To make image palette inspection (Rule V-01) fast and zero-dependency in Round 2, we paired client-side HTML5 Canvas pixel sampling with backend RGB Euclidean distance math.

7. Three Major Improvements Committed for Round 3

Round 3 Improvement	What It Is & What It Changes for the User (Substantial Engineering Scope)
1. Multi-Format Document Ingestion & Layout-Aware Parsing (PDF & DOCX)	What it is: A server-side document ingestion pipeline using pypdf, python-docx, and Pillow that parses uploaded PDF brochures and Word drafts into structured sections, headings, tables, embedded images, and hyperlinks. What it changes for the user: Creators can upload real campaign PDFs and DOCX files directly instead of pasting text, and every finding is anchored to its exact page number, paragraph index, or embedded asset.
2. Semantic Fact & Claim Retrieval via Vector Embeddings	What it is: A semantic vector retrieval layer over the Brand Knowledge Base that embeds approved facts and claims into vector space and computes cosine similarity against candidate sentences before invoking the AI judge. What it changes for the user: Catches paraphrased or indirectly worded factual contradictions (e.g., 'converses across two dozen global tongues' vs. Fact F-12: 'Supports 5 languages') that keyword patterns miss, displaying a semantic similarity score on each finding.
3. Multimodal Vision-Model Brand Asset & Logo Verification	What it is: A multimodal vision-inspection pipeline for uploaded campaign graphics and social banners that verifies logo clear-space ratios, detects stretched or unapproved logo variants, and measures text-over-image contrast across photographic backgrounds. What it changes for the user: Design and marketing teams receive automated visual governance on complete ad creatives, catching logo misuse and illegible text overlays before publication.

8. Prototype Verification Results & Team Member Contributions (Round 2)

Verification Measure	Round 1 Target	Measured Result in Round 2 Working Product
Seeded issues detected (tone, fact, terminology, claim, visual, a11y)	At least 80% recall	100% recall across all seeded violations in Submissions #1 and #2 (verified by automated pytest suite).

Verification Measure	Round 1 Target	Measured Result in Round 2 Working Product
Fact and claim findings backed by a real approved source	100%	100% Evidence Grounding Rate enforced by validate_and_ground_findings() in backend/engine.py.
Confidential text leaked to external AI	0 bytes leaked	0 bytes leaked: 100% of [[double-bracket]] spans are replaced with [REDACTED_CONFIDENTIAL_N] before AI calls.
End-to-end automated integration test suite	Core workflow coverage	5/5 end-to-end pytest tests passing (tests/test_workflow.py) covering all 9 workflow steps.

Team Member	Role on the Project	Part Built in Round 2 (Certificate Contribution)
Logeshwaran V (Team Lead)	Team Lead & Product Architect	Designed the Brand Knowledge Base v3.0 rule schema, the 9-step governance workflow state machine, evidence-grounding architecture, and full-stack system integration.
Muthukumar A	Backend & Hybrid AI Engineer	Built the FastAPI backend, SQLite relational schema, deterministic compliance engine, Claude API + offline heuristic judge, Evidence Validator, and HMAC token authentication.
Mohammed Arshath	Frontend & UX Engineer	Built the Apple-HIG Review Studio SPA, interactive annotated canvas, word-level diff viewer, live WCAG contrast & palette studio, Governance Insights dashboard, and 27-step guided tour.
Poojitha V S	Quality, Data & Security Lead	Built the confidential [[double-bracket]] redaction pipeline, SHA-256 tamper-evident compliance certificates, AI trust & audit log, pre-seeded demo scenarios, and pytest suite.